REVISION 2021 • SEMESTER 6 • FIRST YEAR / COMMON

Major Project

A full-screen, syllabus-style digital handbook for Course Code 6009. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6009

COURSE TITLE
Major Project

DEPARTMENT
First Year / Common

TYPE
Project

SEMESTER
6

CATEGORY
Final Semester Project

USE
Study + notes PDF

FORMAT
Big handbook

6009

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map	**objectives + outcomes**	**Module lessons**	**4 detailed units**
Formula bank	**calculations**	**Practice bank**	**questions + tasks**
Model paper	**official style**	**Answer key**	**full points**

Course objectives and syllabus map

Objectives

- ✓ Select a realistic technical problem and solve it systematically
- ✓ Design, fabricate/test and document a working project
- ✓ Prepare report, presentation, demonstration and viva answers

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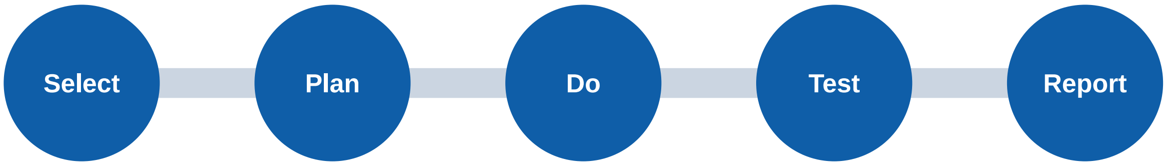
Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Problem selection	Select a problem from industry, society, lab or local need; Define scope, constraints, users and expected output	Short notes, calculations, diagram, checklist, viva answers
M2	Design and planning	Prepare block diagram, workflow, material list and schedule; Estimate cost, risk and responsibilities	Short notes, calculations, diagram, checklist, viva answers
M3	Implementation and testing	Fabricate or develop the solution as per approved design; Test each subsystem before full integration	Short notes, calculations, diagram, checklist, viva answers
M4	Report and presentation	Write abstract, introduction, design, working, testing, cost and conclusion; Include circuit/mechanical drawings, tables and photos	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Problem selection

Core explanation

- ✓ Select a problem from industry, society, lab or local need
- ✓ Define scope, constraints, users and expected output
- ✓ Conduct literature/product survey and compare existing solutions
- ✓ Prepare problem statement and objectives

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

Design and planning

Core explanation

- ✓ Prepare block diagram, workflow, material list and schedule
- ✓ Estimate cost, risk and responsibilities
- ✓ Select components, tools, software and testing method
- ✓ Prepare review-I presentation

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

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- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 3

Implementation and testing

Core explanation

- ✓ Fabricate or develop the solution as per approved design
- ✓ Test each subsystem before full integration
- ✓ Record readings, failures, corrections and improvements
- ✓ Use safety precautions during fabrication and testing

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

Report and presentation

Core explanation

- ✓ Write abstract, introduction, design, working, testing, cost and conclusion
- ✓ Include circuit/mechanical drawings, tables and photos
- ✓ Prepare demonstration checklist and backup plan
- ✓ Answer viva from actual design choices and test results

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

Project Progress % = Completed Milestones / Planned Milestones x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

Cost Variance = Actual Cost - Planned Cost

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

Efficiency = Useful Output / Input x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

Reliability Evidence = Repeated Tests Passed / Total Tests

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Prepare project proposal in one page
- ✓ Create Gantt chart with milestones
- ✓ Maintain test log and correction log
- ✓ Prepare final demo checklist

Important keywords

Problem statement · Prototype · Testing · Report · Costing · Milestone · Viva · Demo

Short questions

Q1.1	Explain: Select a problem from industry, society, lab or local need	Answer with definition, process, application and one example.
Q1.2	Explain: Define scope, constraints, users and expected output	Answer with definition, process, application and one example.
Q1.3	Explain: Conduct literature/product survey and compare existing solutions	Answer with definition, process, application and one example.
Q2.1	Explain: Prepare block diagram, workflow, material list and schedule	Answer with definition, process, application and one example.
Q2.2	Explain: Estimate cost, risk and responsibilities	Answer with definition, process, application and one example.
Q2.3	Explain: Select components, tools, software and testing method	Answer with definition, process, application and one example.
Q3.1	Explain: Fabricate or develop the solution as per approved design	Answer with definition, process, application and one example.
Q3.2	Explain: Test each subsystem before full integration	Answer with definition, process, application and one example.
Q3.3	Explain: Record readings, failures, corrections and improvements	Answer with definition, process, application and one example.

Q4.1	Explain: Write abstract, introduction, design, working, testing, cost and conclusion	Answer with definition, process, application and one example.
Q4.2	Explain: Include circuit/mechanical drawings, tables and photos	Answer with definition, process, application and one example.
Q4.3	Explain: Prepare demonstration checklist and backup plan	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to Problem statement. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define Major Project.
2. List four important keywords: Problem statement, Prototype, Testing, Report.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Problem selection.
7. Compare two important concepts from Design and planning.
8. Explain the practical importance of Implementation and testing in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for Major Project in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of Major Project.
2. Use keywords: Problem statement, Prototype, Testing, Report, Costing, Milestone, Viva, Demo.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Problem selection

- Select a problem from industry, society, lab or local need
- Define scope, constraints, users and expected output
- Conduct literature/product survey and compare existing solutions
- Prepare problem statement and objectives

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

Design and planning

- Prepare block diagram, workflow, material list and schedule
- Estimate cost, risk and responsibilities
- Select components, tools, software and testing method
- Prepare review-I presentation

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Implementation and testing

- Fabricate or develop the solution as per approved design
- Test each subsystem before full integration
- Record readings, failures, corrections and improvements
- Use safety precautions during fabrication and testing

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

Report and presentation

- Write abstract, introduction, design, working, testing, cost and conclusion
- Include circuit/mechanical drawings, tables and photos
- Prepare demonstration checklist and backup plan
- Answer viva from actual design choices and test results

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.